

Vulnerability Assessment

Have You Done it Yet?

Regular vulnerability assessment of your systems/networks is a must. OpenVAS is one such tool that can assist you.

You don't need to read a book to understand why vulnerabilities in systems can put them at potential risks. Vulnerabilities may exist in your network due to misconfigured or unpatched systems. This brings us to the subject of vulnerability assessment and management—a process of identifying the presence of vulnerabilities and reducing the security risk of your system to an acceptable level.

There are various assessment tools that assist IT security professionals in identifying vulnerabilities in servers and desktops. You can initiate a scan and identify potential network and system weaknesses using vulnerability scanners. These security scans can run either on remote or local systems against target systems.

Challenges in the current scenario are:

1. The non-availability of open source vulnerability assessment tools
2. Understanding the architecture of VAS (vulnerability assessment systems)
3. Implementing a VAS tool.

VAS architecture

Typically, the design of a VAS should meet the following basic requirements:

1. Configured to reach all target systems
2. Authenticated access to the VAS server and sessions
3. A constant updating of the VAS plug-in database before initiating a scan
4. Defining a scan frequency and ear-marking

the target systems that fall within the scope of scanning

5. Ensuring secure availability of scan reports with time stamping.

OpenVAS—Open Vulnerability Assessment Scanner, a fork of the now closed-source Nessus security scanner—is an open source vulnerability assessment tool. It works on a client-server architecture, which has the following four components:

1. OpenVAS server
2. OpenVAS feed server
3. OpenVAS client
4. Target systems

The OpenVAS server requires to be configured in a manner to fetch updates or the latest plug-in feeds from the publicly-hosted OpenVAS Feed Repository server. The OpenVAS server will contact the repository server to identify available updates and download them.

The OpenVAS client is a GUI client used to manage, configure and report scan results. Access to the OpenVAS server is controlled by a secure password- or certificate-based authentication. Through the OpenVAS client software you can specify target systems, port range, NVT (network vulnerability test) and initiate the scan. The OpenVAS client will show you the real-time status of your scan and report the relevant results. You can manage various scans by grouping IP addresses based on category and saving them in separate sessions. Results under each session are stored in that particular session